

Assignment 1 Solutions

Fall 2026

$$1.2 \begin{bmatrix} 1 & 8 & 7 & -1 \\ 4 & -9 & 0 & 8 \\ 7 & -7 & -3 & 10 \end{bmatrix}$$

augmented

$$\begin{bmatrix} 1 & 8 & 7 \\ 4 & -9 & 0 \\ 7 & -7 & -3 \end{bmatrix}$$

coeff.

$$1.5 \begin{aligned} 1 - 2(3) + 2 - 2(3) &= 1 - 6 + 2 - 6 = -9 \checkmark \\ 1 - 3 - 2 - 2(3) &= 1 - 3 - 2 - 6 = -10 \checkmark \\ -3(1) + 8(3) - 6(2) + 4(3) &= -3 + 24 - 12 + 12 = 21 \checkmark \\ 2(3) - 7(2) + 7(3) &= 6 - 14 + 21 = 13 \checkmark \end{aligned}$$

$$1.7 \begin{bmatrix} 3 & 6 & -15 \\ -1 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -5 \\ -1 & 1 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -5 \\ 0 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -5 \\ 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \end{bmatrix}$$

$R_1 \leftarrow R_1/3 \quad R_2 \leftarrow R_2 + R_1 \quad R_2 \leftarrow R_2/3 \quad R_1 \leftarrow R_1 - 2R_2$

$$\begin{aligned} x_1 &= -5 \\ x_2 &= 0 \end{aligned}$$

$$1.12 \begin{bmatrix} -3 & 6 & 0 & 9 \\ 3 & -8 & 0 & -9 \\ 0 & 2 & 6 & 18 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -3 \\ 3 & -8 & 0 & -9 \\ 0 & 1 & 3 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & -2 & 0 & 0 \\ 0 & 1 & 3 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 9 \end{bmatrix}$$

$R_1 \leftarrow R_1/-3 \quad R_3 \leftarrow R_3/2 \quad R_2 \leftarrow R_2 - 3R_1 \quad R_3 \leftarrow R_3/3 \quad R_1 \leftarrow R_1 - 2R_2$

$$\sim \begin{bmatrix} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$R_3 \leftarrow R_3/3 \quad R_1 \leftarrow R_1 + 2R_3$

$$x_1 = -3$$

$$x_2 = 0$$

$$x_3 = 3$$

$$1.13 \quad \begin{bmatrix} 1 & -2 & -2 & -7 \\ -1 & 3 & 2 & 10 \\ 2 & -6 & -3 & -18 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & -7 \\ 0 & 1 & 0 & 3 \\ 2 & -6 & -3 & -18 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & -7 \\ 0 & 1 & 0 & 3 \\ 0 & -2 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & -7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$R_2 \leftarrow R_2 + R_1$ $R_3 \leftarrow R_3 - 2R_1$ $R_3 \leftarrow R_3 + 2R_2$ $R_1 \leftarrow R_1 + 2R_3$ $R_1 \leftarrow R_1 + 2R_3$

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \quad \begin{array}{l} x_1 = 3 \\ x_2 = 3 \\ x_3 = 2 \end{array}$$

we know it
has a unique solution
at this point

$$1.16 \quad \begin{bmatrix} -4 & 1 & 3 \\ -7 & 0 & 5 \\ -4 & 3 & 1 \end{bmatrix} \sim \begin{bmatrix} -4 & 1 & 3 \\ -7 & 0 & 5 \\ 17 & 3 & -14 \end{bmatrix} \sim \begin{bmatrix} -4 & 1 & 3 \\ -7 & 0 & 5 \\ 5 & 6 & -5 \end{bmatrix} \sim \begin{bmatrix} 5 & 6 & -5 \\ -7 & 0 & 5 \\ -4 & 1 & 3 \end{bmatrix}$$

$R_2 \leftarrow R_2 - 3R_1$ $R_3 \leftarrow R_3 + 3R_1$ $R_1 \leftrightarrow R_3$

$$1.22 \quad x = -2y \quad \begin{array}{l} x = 0 \\ y = 0 \end{array}, \quad \begin{array}{l} x = 1 \\ y = -\frac{1}{2} \end{array}, \quad \begin{array}{l} x = -2 \\ y = 1 \end{array}$$

$$1.25 \quad \begin{array}{l} x_1 = 1 - x_2 + 3x_4 + 3x_7 \\ x_3 = 3 + x_4 - 3x_6 - 2x_7 \\ x_5 = -3 - 3x_6 + 4x_7 \end{array}$$

$$\begin{array}{l} x_1 = 1 \\ x_2 = 0 \\ x_3 = 3 \\ x_4 = 0 \\ x_5 = -3 \\ x_6 = 0 \\ x_7 = 0 \end{array}$$

$$\begin{array}{l} x_1 = 0 \\ x_2 = 1 \\ x_3 = 3 \\ x_4 = 0 \\ x_5 = -3 \\ x_6 = 0 \\ x_7 = 0 \end{array}$$

$$\begin{array}{l} x_1 = 4 \\ x_2 = 0 \\ x_3 = 4 \\ x_4 = 1 \\ x_5 = -3 \\ x_6 = 0 \\ x_7 = 0 \end{array}$$

1.42 True. We showed this, it is the basis of row reduction

1.43 False. We showed a linear system has 0, 1, or ∞ solutions.

1.44 True. Row equivalent matrices have the same solutions

$$1.51 \text{ False.} \quad \begin{array}{l} x + y + z = 1 \\ x + y + z = 2 \end{array}$$

1.57 False. $x + y = 2$
 $2x + 2y = 4$

1.65 The xy -plane is given by the equation $z = 0$.

So we're interested in the point (x, y, z) s.t.

$$2x + 3y + 3z = 6 \text{ and } z = 0, \text{ i.e., } 2x + 3y = 6.$$

This has slope-int. form $y = -\frac{2}{3}x + 2$

We can also think of this as solving the system with aug. matrix:

$$\begin{bmatrix} 2 & 3 & 3 & 6 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

1.66 $\begin{bmatrix} 1 & 4 & -1 \\ -3 & -h & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 4 & -1 \\ 0 & -(h+12) & 10 \end{bmatrix}$

For this echelon form to be inconsistent it must be that $-(h+12) = 0$

that is $h = -12$. Another way to think about it: we need the two equations to be parallel lines.

There is no h for which there are infinitely many solutions.

A pair of lines has ∞ simultaneous solutions only if they represent the same line. This is not possible here.